

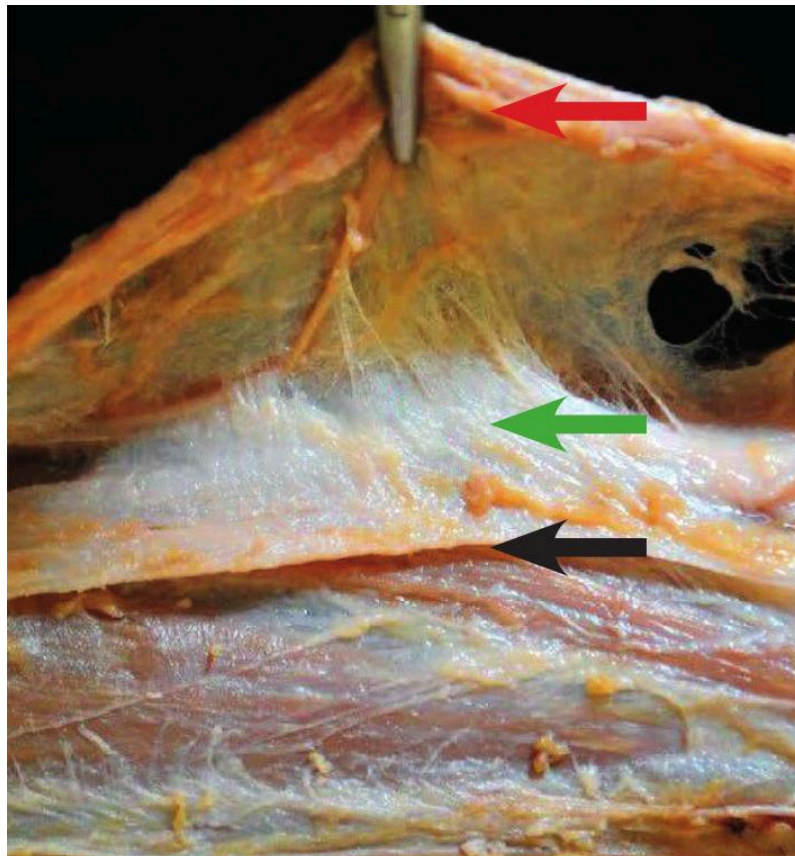
FASCIA, MUSCLE CHAINS AND BIOTENSEGRITY

FASCIAL ANATOMY: SUPERFICIAL FASCIA AND DEEP FASCIA

Fascia: a mysterious substance that permeates the human body. A continuous, interconnected structure with lots of different names. Fascia is connective tissue. It surrounds, encloses, supports and connects each and every element of the human body. Fascia *is* connection. On a microscopic level, it's a liquid tissue (non-Newtonian fluid). Its key functions are to transfer energy and to help other structures glide over each other by reducing excessive wear or friction that could result in injury.

Fascia can be divided into two main groups:

- *Superficial fascia*: this is located beneath the dermis and is made up of loose connective tissue and varying amounts of fat. It's like the thin silvery film on a joint of meat at the butcher.
- *Deep fascia*: this is connected to superficial fascia and is made up of dense connective tissue that surrounds and sections off different structures. It connects absolutely everything in the human body: from the muscles in our feet to the vital organs in the abdomen, the nerves in the cervical spine, and so on, linking everything to everything.



SUPERFICIAL FASCIA OF THE FOREARM AROUND THE EXTENSOR MUSCLES. THE RED ARROW INDICATES THE SUPERFICIAL FASCIA; THE GREEN ARROW INDICATES CONNECTIVE TISSUE MADE FROM COLLAGEN; AND THE BLACK ARROW INDICATES THE INTERMUSCULAR SEPTUM (DONES ET AL., 2013)



DEEP FASCIA PHOTOS: JULIAN BAKER

Interestingly, unless the fascia cells (known as fibroblasts) are stimulated directly by way of myofascial release, they are very slow to adapt to movement, unlike our muscles, which respond instantly. But what does this mean and how is it connected to climbing? In practice, it has a big impact on climbing performance. Imagine you've been sitting at a desk for the past eight hours; you then arrive at the wall and start training like mad without warming up or doing any myofascial release. Although your body is already climbing, the fibroblasts in your fascia are still 'sitting down', in the same position they've been in all day. They haven't had time to adapt to this change in movement. Then one day, you'll be climbing away and suddenly feel a strange tightness or strain somewhere for 'no reason at all' ...

This is why we should include a good dose of myofascial release in the mobility section of our warm-up, either by flossing, using massage balls or skin rolling (ask your physio to show you how). These techniques target the superficial fascia which, in turn, helps to release the deep fascia. As a result, muscle fibres will glide more smoothly over each other, any tension built up over the day will be reduced and you'll see better quality of movement.

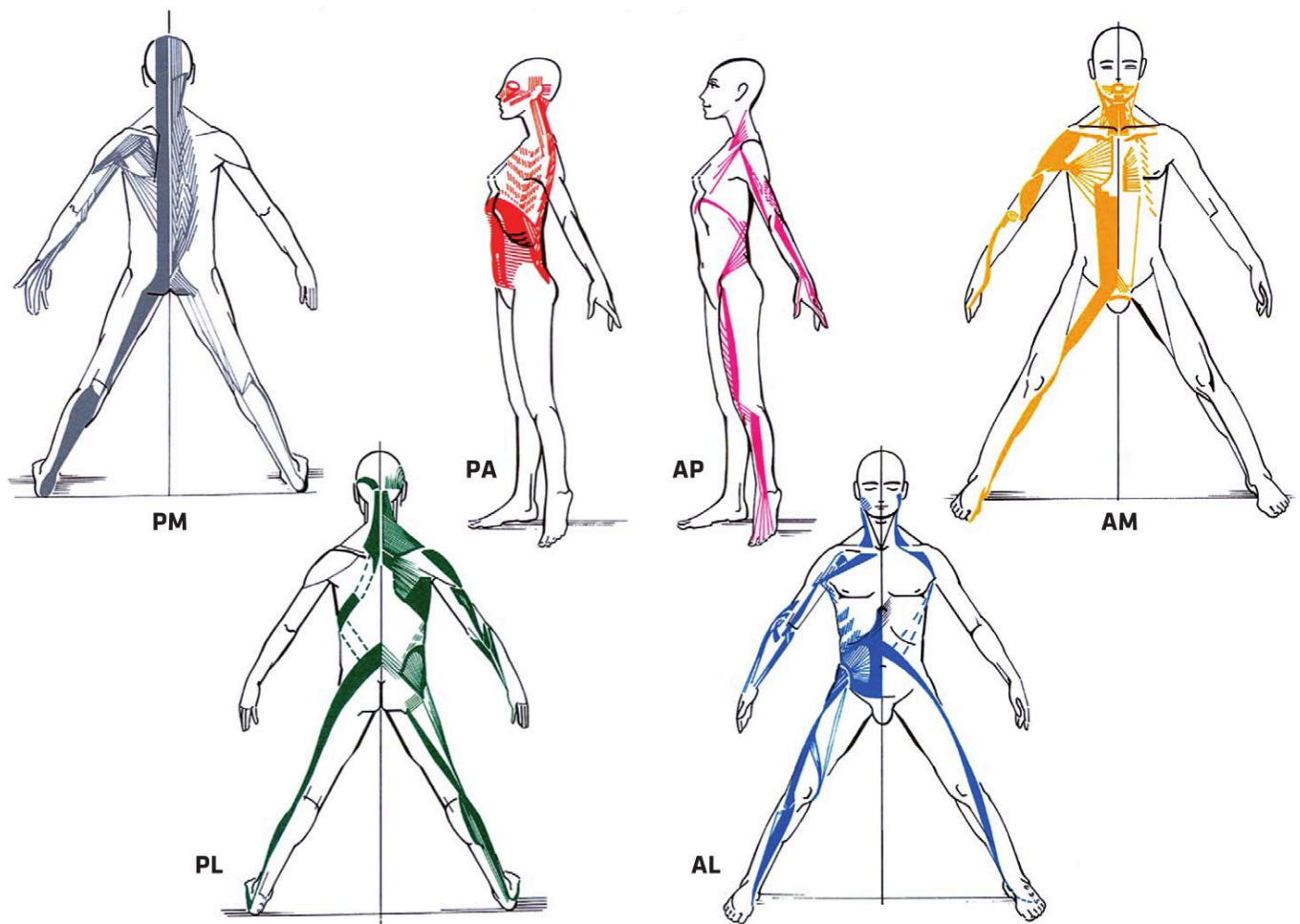
MAIN MUSCLE CHAINS

Many authors (such as Busquet, Kabat and Mézières) have studied different systems of muscle chains and each has developed their own way of explaining the connectivity and function of our muscles, with pros and cons to each model. Personally, I work with the GDS system designed by Denys-Struyf (2008), which describes six different muscle chains (see [image on following page](#)):

- *PM*: postero-median chain. The extensor chain that saw us evolve from walking on four limbs to walking on two. It mainly runs down the back of the body.
- *AM*: antero-median chain. The main antagonist chain to the PM muscles. It is a closing and flexion chain and mainly runs down the front of the body.
- *PA and AP, sometimes referred to as PAAP*: postero-anterior and antero-posterior chains. Two complementary chains that can function as a single chain. They contain muscles deep within the body and are responsible for joint adjustment.
- *AL*: antero-lateral chain. A chain that criss-crosses through the entire body and is mainly responsible for closing movements and internal rotation.
- *PL*: postero-lateral chain. A chain that criss-crosses through the entire body and is mainly responsible for opening movements and external rotation.

Why use the GDS system? Denys-Struyf studied these chains and the postural changes caused by overloading the muscles in each chain in bipeds in static positions. These excessive loads modify joint alignment, reducing efficiency and causing significant wear. They also create a tendency to overload certain muscles and increase the risk of certain injuries.

The complex system of GDS muscle chains is a brilliant tool for injury prevention, among many other things. As such, if at all possible, I'd recommend finding a physio or a coach who works with this method. They can do an assessment and show you preventative exercises to address any imbalances within your muscle chains.



MUSCLE AND JOINT CHAINS ACCORDING TO DENYS-STRUYF

THE BODY AS A BIOTENSEGRITY STRUCTURE

Biotensegrity: another unusual word. Simply put, it describes the human body as a structure in a state of balanced tension. Any change in tension or position needs to be compensated for in another part of the body. The concept complements, expands on and supports the ideas discussed in this chapter.

On a practical level, if an excessive load in the forearm isn't stopped and compensated for in time, it will need to be compensated for elsewhere – for example, in the upper arm, the shoulder or the neck.

Following this rather abstract section, which you might think has nothing to do with training for climbing, I'd urge you to take away one simple idea: the body as a single, interconnected being. It's yours and you're part of it: listen to it and look after it. It isn't something to be trained or punished. If you look after it to the best of your ability, it will allow you to perform to the best of your ability, responding how and when you want it to. And if your body is trying to tell you something, don't ignore it: respect it.